An Essay About Science, or What Is Science Anyway?

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We live in a technological world; the environment in which we are immersed is largely a byproduct of the advances in what we call "scientific knowledge" (especially the rapid developments of the last 100 years).

But what is science? Science is nothing more than a way of looking at the universe that begins by assuming that there exists such a thing as an objective reality external to each individual—that is, a reality independent of the observer and of their prejudices. This condition makes science something very different from any other way of viewing the world. Non-abstract art, for example, provides us with representations of our surroundings that can come to involve, quite dramatically, the most intimate emotions not only of the creator of a work but also of those who contemplate it. This brief discussion aims to make explicit a few of the elements that constitute the signature of the way science represents the world.

A representation of reality. Paul Cézanne, *Mont Sainte-Victoire* (1905)

We have always felt a deep curiosity about nature; we have marveled at it, and from that have emerged ideas that we have later put to our service. A very clear example of this

is found in astronomical observations which eventually became basic tools for agriculture and navigation—both fundamental technologies for the advancement of civilization. In their beginnings, observations of the sky were surely contemplative. There were the daytime and nighttime skies, the sun, the moon, and the stars, and these could not but inspire wonder; it was worth observing and asking: what is it that we see? The initial answers to those questions were most likely naïve and lacking in order, but repeated observation and the exchange of ideas must have led to an organization of those first answers, turning them into the basic elements of different mythologies. These mythologies developed into their final forms only when there was a sufficient degree of systematization of observations of the sky. Such systematization led, among other things, to the discovery of the relationship between the positions of the sun in the sky (solstices and equinoxes) and seasonal changes—knowledge fundamental to agriculture, which in turn is one of the basic technologies enabling the transition from nomadic or migratory civilizations to sedentary ones.

Over time, observations of nature and the ideas explaining those observations evolved much further. Physics, for example, reached a very high point with Aristotle (384 BC – 322 BC), whose ideas remained virtually unchanged until the emergence of the works of Galileo (1564–1642) and Newton (1642–1727), which mark the beginning of physics and science as we understand them today—that is, as disciplines of modern science.

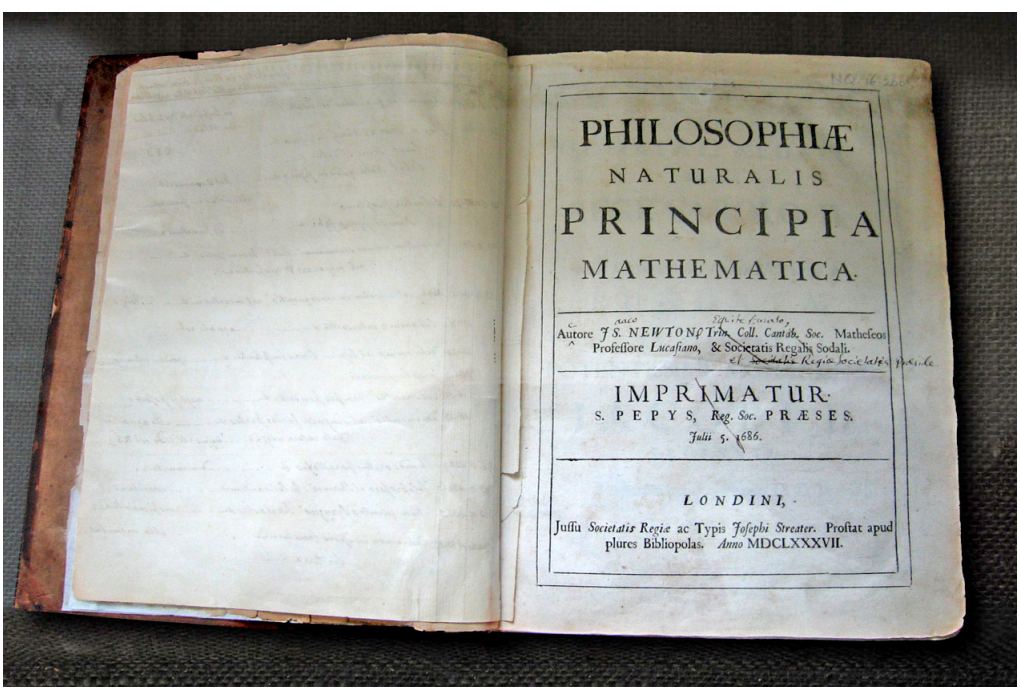
But let us return to our initial question: what is science? This is very difficult to define precisely, so it is probably best to limit ourselves to some characteristic aspects of scientific knowledge that distinguish it from any other way of representing the world.

A first element of the signature of science is its objectivity; in order to achieve it, science uses a common language: mathematics, and consequently limits the elements of reality of interest to those that can be quantified. Such elements of reality are described in terms of (numerical) experimental measurements, from which a property called "reproducibility" is required. This is very simply explained: if the same experiment is carried out by different groups of practitioners of science under similar circumstances, the results must agree.

The use of measurements introduces a second fundamental element in the character of science: the inevitable existence of so-called "experimental errors," which is worth discussing through an example. Suppose we wish to determine the height of a child. This can be done in various ways: with the child standing or lying down, and using different instruments, such as a ruler or a flexible measuring tape. To simplify the problem somewhat, suppose we ask several people to carry out the measurement using the same method and exactly the same type of instrument. What will most likely happen is the following: several results will be found—perhaps one measurement yields a height of 1.56 m, another 1.55 m, and a third 1.54 m. These values give rise to an obvious question: which of the three corresponds to

the child's true height? The scientific answer is the following: the child's height is 155 cm with an uncertainty of 1 cm—that is, the child's height lies between 154 and 156 cm. The uncertainty of 1 cm is called experimental error, and the fact that the three values are close to each other within an amount less than or equal to the measurement error is interpreted as reproducibility. If we had measured the child's height a large number of times, we would expect most measurements to be quite close to 155 cm with an error smaller than 1 cm; a value of, say, 165 cm would deviate from the expected value by 10 cm, which is significantly larger than the error and is interpreted as an anomaly that must be investigated in detail.

Scientific knowledge is much more than just a vast collection of measurement results. Our observations must be classified and organized in a certain way in order to constitute science; this quality—organization—is another characteristic element of science. From time to time, data from observations of different aspects of a phenomenon (or a set of phenomena) can be connected through non-random mathematical relationships; with some luck, a set of such relationships can be placed within a general framework, and that framework constitutes a scientific theory (a structure of thought). Newton's laws of mechanics constitute a paradigmatic example of a scientific theory: in its three postulates and a few implicit assumptions, Newton's laws express everything we need to describe any phenomenon associated with motion (of macroscopic bodies moving at speeds small compared to that of light).


PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA.

Autore JS. NEWTONO Trin. Coll. Cantab. Soc. Matheseos Professore Lucasiano, & Societatis Regalis Sodali.

IMPRIMATUR.
S. PEPYS, Reg. Soc. PRÆSES.
Julii 5. 1686.

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Jussu Societatis Regiæ ac Typis Josephi Streater. Prostat apud plures Bibliopolas. Anno MDCLXXXVII.

Newton's *Principia* (1687), a true scientific paradigm.

From scientific theories we also require something extremely strong: a predictive character; we expect them to be capable of predicting phenomena not observed in the experiments that gave rise to the theory in question. An emblematic example helps us understand what we mean by the predictive character of science: the discovery of the planet Neptune. Until the first half of the 19th century, astronomers knew of the existence of seven planets in the solar system. The motions of six of these planets followed quite rigorously the laws of Newtonian mechanics and the law of universal gravitation; Uranus, however, showed certain anomalies in its orbital parameters. A firm belief in the theory led John Couch Adams in 1843, and independently Urbain Le Verrier in 1846, to apply Newton's laws to try to explain Uranus's orbital anomalies. The work of these two men led not merely to the prediction of the existence of an eighth planet, but to the calculation of its position in the sky; on September 23, 1846, James Challis observed Neptune precisely in the position predicted by Adams and Le Verrier—a true and remarkable triumph of theory.

It is interesting to note that a given set of observations may be explained by several different theories; the choice of the dominant (or correct) theory lies in the number of observations it can explain. In general, the theory accepted as correct is the most successful in the sense of explaining the greatest number of observed phenomena. Now, and this is another fundamental characteristic of science, theories are perishable: we accept them until they reach a stage of obsolescence, typically characterized by the appearance of new phenomena that the theory cannot explain. Quantum mechanics provides a representative example of the replacement of one theory by another. Toward the end of the 19th century, there was a sense that nothing new remained to be explained: classical mechanics, electromagnetism, and thermodynamics were the three theories that accounted for all phenomena. Around that same time, however, a certain problem began to be studied (the microscopic thermodynamics of electromagnetic radiation contained in a cavity). The result of these studies was devastating, to say the least, as the theory exhibited enormous inconsistencies (it predicted that the energy contained in the cavity should be infinite), and physics thus found itself at an impasse. On December 14, 1900, Max Planck presented a paper to the Berlin Academy of Sciences that resolved the paradox through the introduction of a new idea: the so-called quantum hypothesis. During the first quarter of the 20th century, both theoretical and experimental research focused on a multitude of phenomena, and by the end of 1926 (Copenhagen), quantum mechanics took its modern form with the formulation of its orthodox interpretation.

From the standpoint of worldview, quantum mechanics represents a dramatic shift: it confronts us with the need to admit that all our everyday concepts—position, motion, trajectory—are misleading; they simply lack meaning. There is no reasonable way to speak of the trajectory of an electron; such a concept must be declared nonexistent. The quantum

view of the world is entirely foreign to our most basic intuition. The concept of particle does not exist, nor does that of wave; particles and waves constitute inseparable entities, in apparent violation of the most elementary ideas about such objects. Quantum mechanics simply destroys a century-long "honeymoon" with the ideas of Galileo and Newton—ideas that, however, are recovered from the theory as appropriate mathematical limits.

Two of the basic questions in the philosophy of science are the following: how does scientific knowledge advance? In this discussion we will adhere to the views of Karl Popper (1902–1994). Expressed in the least technical terms possible, we can say that science advances in fits and starts: each time we manage to formulate a theory, we only have a paradigm that explains some things. A theory may successfully predict new phenomena, but this does not make it true; every theory is fallible by its very nature, and science must undertake titanic efforts to seek "critical" experiments—these are simply experiments that reveal phenomena the theory cannot explain, thereby exposing its false elements. In other words, science advances not because of its successes, but by virtue of its failures.

Scientific knowledge is under constant revision. One might say that those who practice science live in a somewhat peculiar state: at all times they are aware that their ideas about the world are very likely wrong, and that sooner or later they will become obsolete and be replaced by entirely new ideas. This is the fundamental element of the signature of scientific knowledge: to remain in a perpetual state of revision.

Science provides us with a particular way of seeing and describing the world; scientific knowledge is self-critical, and this fundamental characteristic makes it very different from the knowledge we use in everyday life to relate to our surroundings and ourselves. This self-criticism makes science dynamic. Common knowledge, on the other hand, is static; it suffers from a perpetual lethargy and considers itself perfectly true. It seems that humans require a certain sense of absolute certainty—perhaps this is an evolutionary adaptation, some biological need to feel secure. In the vast majority of cases, common knowledge is false, as it is based on obvious fallacies: false induction, populist sophisms, appeals to authority—these are some examples of the types of reasoning errors we fall into when attempting to describe observed phenomena. These errors are not absent in science, since science is ultimately a human activity; however, non-scientific knowledge clings fiercely to itself and refuses to acknowledge its errors.

Perhaps it would be beneficial for humanity to incorporate some of the awareness of error that guides science; history is full of tragedies associated with absolute certainty. Genocides, for example, are nothing more than the extreme and direct consequence of a group in power having absolute certainty that it possesses the truth.